

Towards a BCI-based marker of conscious processing during anesthesia

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Workshop on *Exploring EEG and BCI Approaches for Tracking Consciousness*

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Intraoperative awareness still occurs...

"I felt my throat being cut, right to left and calculating the inches to endure... The pain in my body was so extreme."^[1]

- Intraoperative awareness can occur despite apparently adequate anesthesia^[2].
- Awareness still occurs with current monitors^[3-7], indices were developed from adult cohorts^[8,9], are less accurate with age^[10], might be biased by EMG activity^[11, 12]



[1] Leslie et al, Anesth Analg, 2010
 [2] Sandin et al, Lancet, 2000
 [3] Sebel et al. Anesth Analg, 2004
 [4] Myles et al. Lancet, 2004

[5] Mashour et al, Anesthesiology, 2012
 [6] Tasbihgou et al, Anaesthesia, 2018
 [7] Laferrière-Langlois et al, Anesthesia & Analgesia, 2024
 [8] Sigl & Chamoun, J Clin Monitoring, 1994

[9] Drover et al, Anesthesiology, 2002
 [10] Bruhn et al. Anesth, 2000
 [11] Aho et al. Acta Anaesthesiol Scan, 2012
 [12] Schuller et al, British Journal of Anaesth, 2015

Unconsciousness under anesthesia



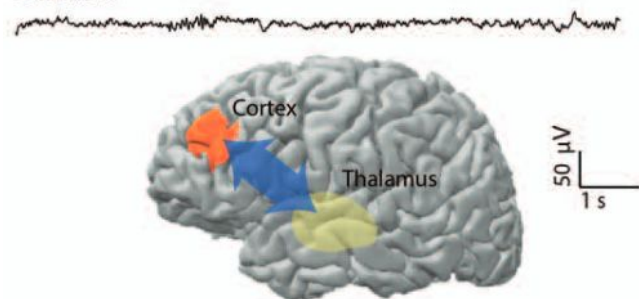
Propofol:

- impairs thalamocortical communication
- disrupts intracortical communication
- alters large-scale cortical dynamics^[1-3]

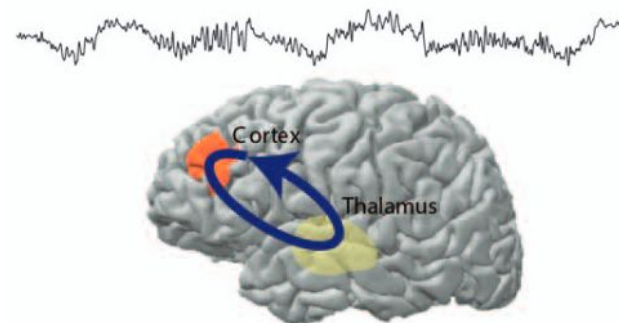
Could an evoked cortical response reveal this disruption?

Thalamocortical Circuits Underlying Propofol-Induced Alpha Oscillations

Conscious



Unconscious



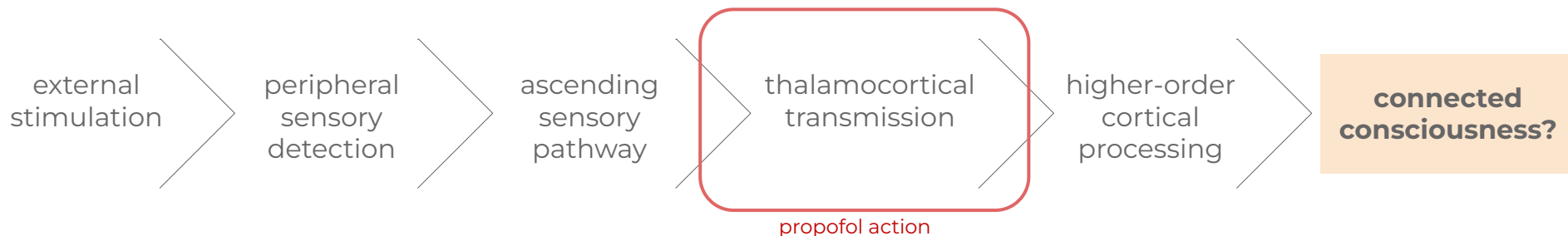
[1] Purdon et al, Anesth, 2015

[2] Brown et al, Annual Rev Neurosc, 2011

[3] Ching et al, PNAS, 2010

Why might an evoked response inform consciousness?

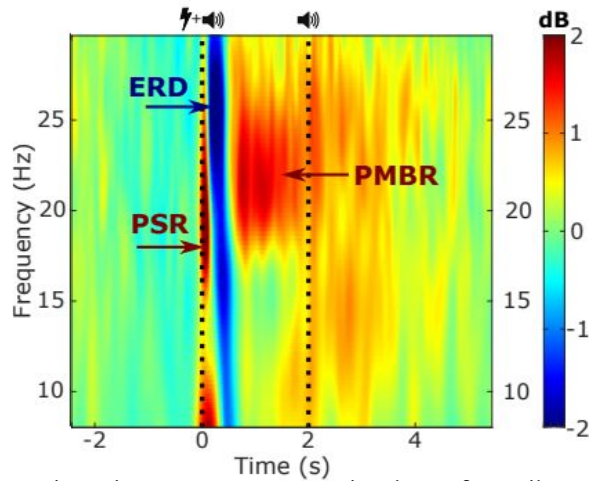
- Current monitors observe spontaneous EEG, but consciousness involves processing of the external world^[1-4]
- Theoretical frameworks distinguish:
 - *dysanaesthesia*^[3]: sensory input may be processed without conscious perception
 - *connectedness*^[4]: conscious perception of the external environment



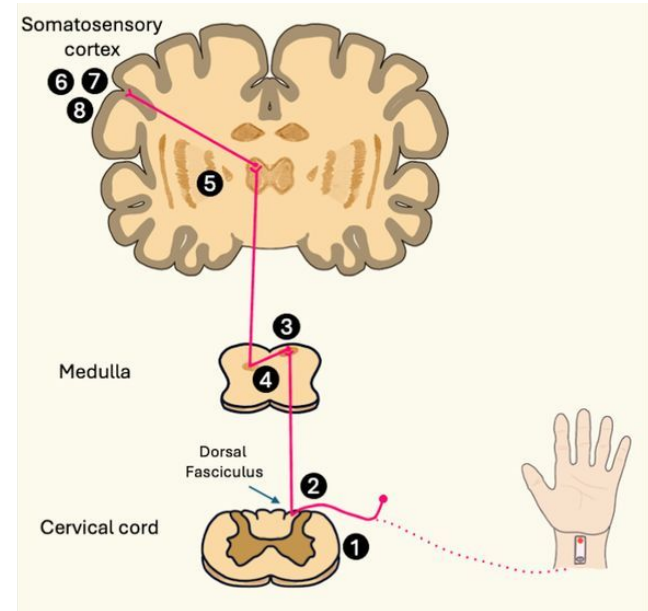
[1] Dehaene & Changeux, Neuron, 2011
[2] Mashour, in The Neurology of Consciousness, 2016
[3] Pandit, Consciousness and Cognition, 2014
[4] Sanders et al, Anesth, 2012

Median Nerve Stimulation (MNS)

- **Painless stimulation** of the median nerve, with an intensity between 8 and 15 mA.
- Instead of passively observing spontaneous EEG, we introduce a **controlled perturbation** and measure the **resulting cortical response**.



Event-related spectrum perturbation of Median Nerve Stimulation (31 subjects)^[1]



Calma et al, Practical Neurology, 2025

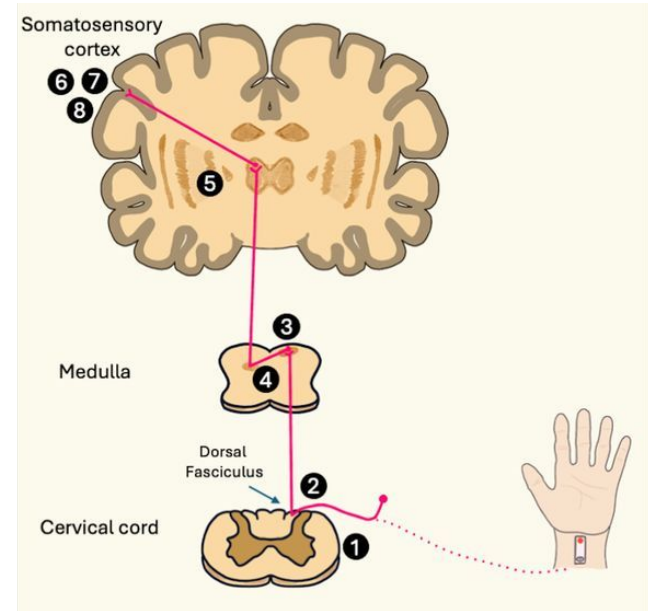
PSR: Post-Stimulation Rebound
ERD: Event Related Desynchronization
PMBR: Post-Movement Beta Rebound

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Hypothesis:

- If conscious processing is **disrupted**, the response should **progressively diverge from the awake reference**.
- If conscious processing is **preserved**, the MNS-evoked response should **remain close to the awake state**.



Calma et al, Practical Neurology, 2025

PSR: Post-Stimulation Rebound
ERD: Event Related Desynchronization
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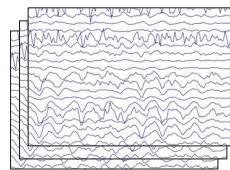
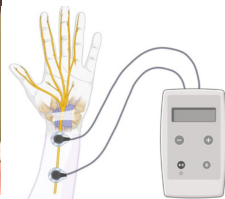
MNS-based BCI to detect deviations from the awake response



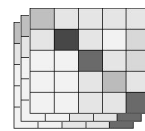
TRAINING PHASE - before the surgery



Recording brain activity with EEG while stimulating the median nerve^[1]



Preprocessing



Feature extraction
(covariance matrices)

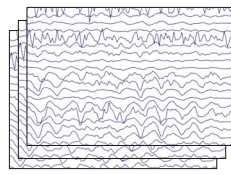
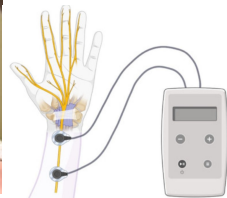


Training the algorithm:
*One-Class Riemannian
Minimum Distance to
the Mean*^[2]

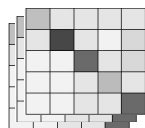
TESTING PHASE - during the surgery



Recording brain activity with EEG while stimulating the median nerve^[1]

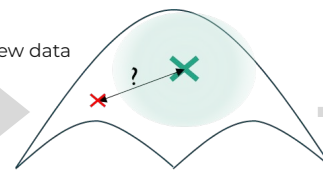


Preprocessing



Feature extraction
(covariance matrices)

new data



Classification with
*One-Class Riemannian
Minimum Distance to
the Mean*

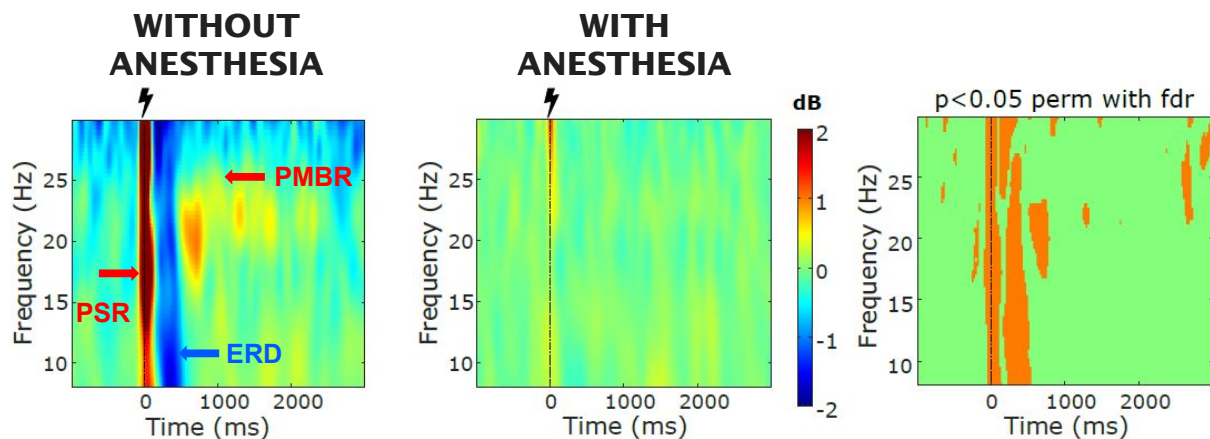
close to the
awake state?

[1] Rimbert et al. JMIR Res. Protoc, 2023

[2] Marissens Cueva et al, 5th Int. Neuroerg Conf 2024

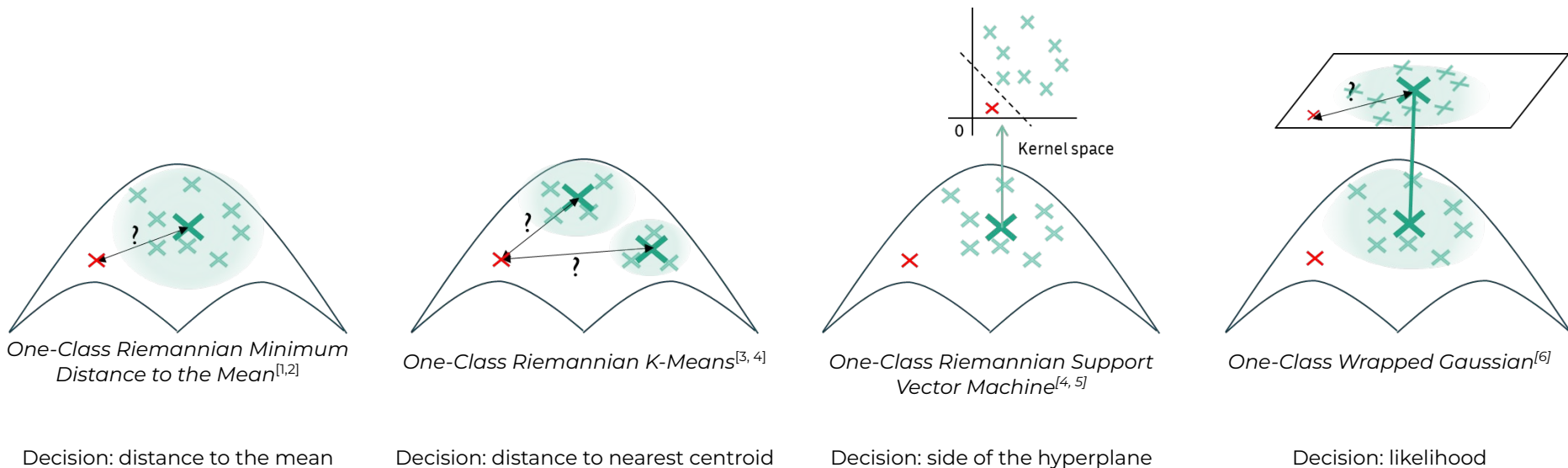
Median Nerve Stimulation pattern under general anesthesia

Grand average time frequency analysis for electrode C4^[1]:



PSR: Post-Stimulation Rebound
ERD: Event Related Desynchronization
PMBR: Post-Movement Beta Rebound

Comparison between different algorithms trained only with the *awake* class



Marissens Cueva et al, *Journal in preparation*



Thursday
September 17
Poster n° 34

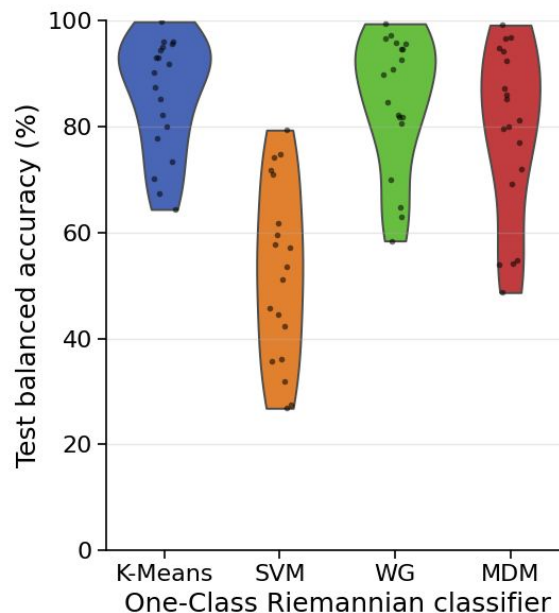
[1] Barachant et al. IEEE Trans. Biomed. Eng. 2012.
[2] **Marissens Cueva** et al, 5th Int. Neuroerg Conf 2024

[3] Navarro-Sune et al, IEEE Trans. Biomed. Eng., 2017
[4] **Marissens Cueva** et al, 11th BCI Meeting, 2025

[5] Schölkopf et al, NIPS., 1999
[6] **Marissens Cueva**, de Surrél, et al, 10th Intern Graz BCI Conf, 2026

Comparison between different algorithms trained only with the *awake* class

Test balanced accuracies of **OC classifiers trained on data when patients are awake** (19 subjects). Best results with OC-RKM are with 2 prototypes, and thresholds of median + 3 × median absolute deviation. For OC-RSVM, $v = 0.5$. *p-value < 0.05

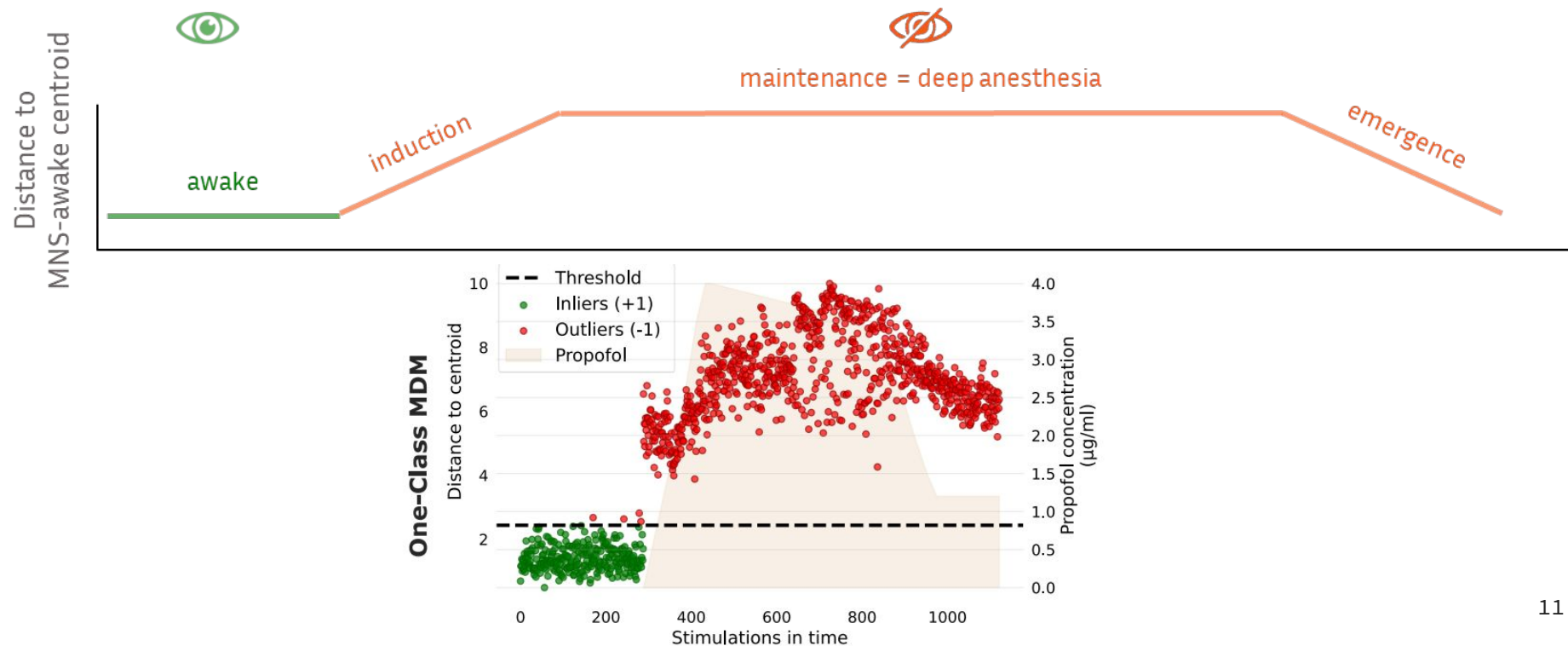


With ϕ the Riemannian kernel:

$$K_{i,j} = \text{tr}(\log(C_{ref}^{-1/2} X_i C_{ref}^{-1/2}) \cdot \log(C_{ref}^{-1/2} Y_j C_{ref}^{-1/2}))$$

Riemannian version of
Support vector data description^[1]

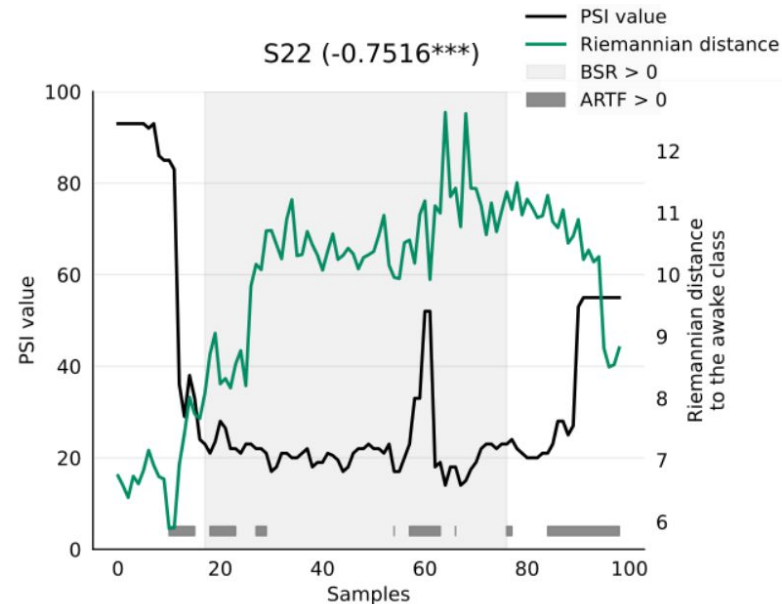
Can the BCI detect deviations from the awake state?



Evolution of the Riemannian distance

9/14 patients: negative correlation with PSI
(increasing deviation from the awake reference with deeper anesthesia)

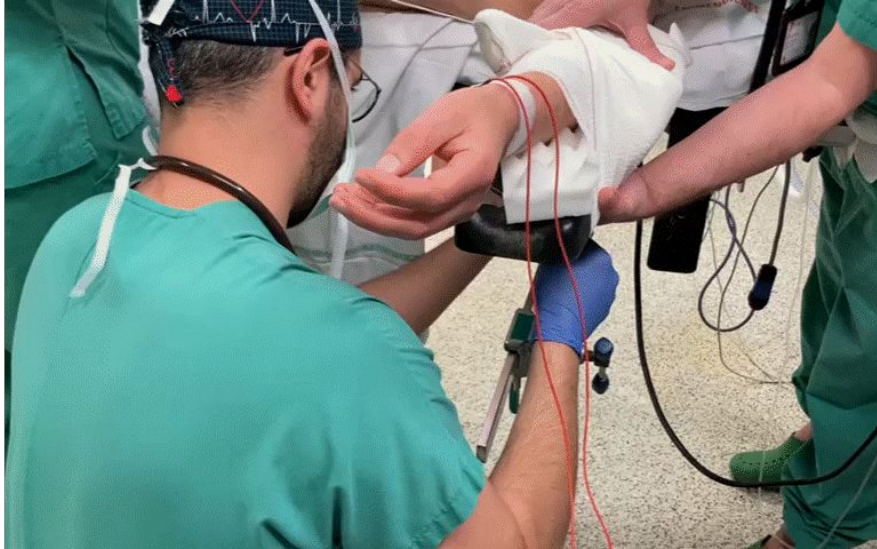
Sedline monitor



PSI: Patient State Index

- 100: fully awake
- 25-50: optimal anesthetic level
- 0: total suppression of brain electrical activity

Clinical validation: two protocols



44 patients (27 females, 52 ± 13 years old)

- Neuromuscular blocking agents:
 - 24 with
 - 20 without
- Surgeries: 15 digestive, 4 gynecology, 16 orthopedics, 3 plastic surgeries, 4 urology, 2 vascular
- Propofol targets:
 - 20 patients ≤ 4 $\mu\text{g/ml}$
 - 18 patients between 4 and 6 $\mu\text{g/ml}$
 - 6 patients between 8 and 11 $\mu\text{g/ml}$

EEG recording in a clinical environment

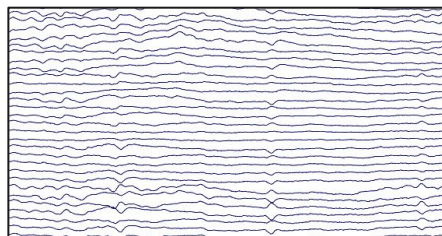
Variability:

- chaotic operation room
- position during surgery
- patient-specific conditions
- opioids

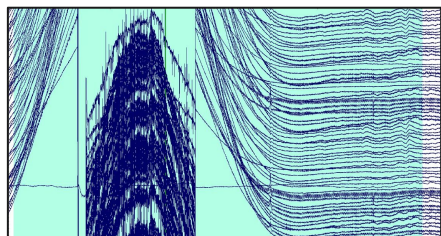


Interferences:

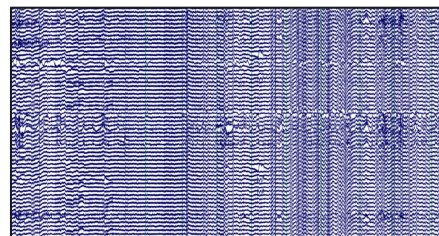
- hot air blankets^[1]
- electrocautery^[2]
- individual socket
- line noise (50 Hz or 60 Hz)
- movement



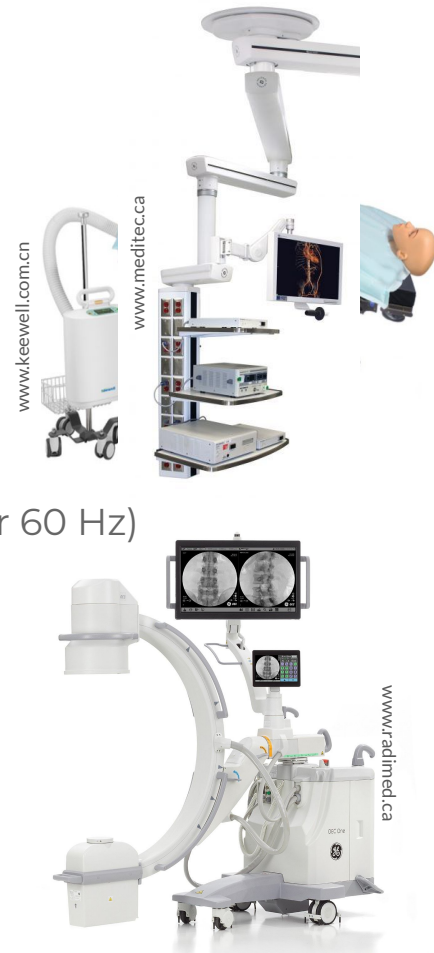
CLEAN



ELECTROCAUTERY



LINE



[1] Hemmerling et al. Anesth. Analg. 2002.

[2] Hight et al. Can. J. Anesth. 2021

Differences with commercial monitors

	COMMERCIAL MONITORS (EX: PSI, BIS...)	OUR APPROACH
Stage	Commercialization ^[1,2]	Research
EEG	Spontaneous	MNS-evoked
Reference	Population-based ^[3]	Patient-specific awake reference
Output	Depth-of-anesthesia index	Deviation from awake response
Algorithm	Pre-trained & proprietary ^[1,2]	One-class learning
Electrodes	Frontal	Frontal & sensorimotor

[1] Sigl & Chamoun, J Clin Monitoring, 1994

[2] Drover et al, Anesthesiology, 2002

[3] Bruhn et al, BJA, 2006

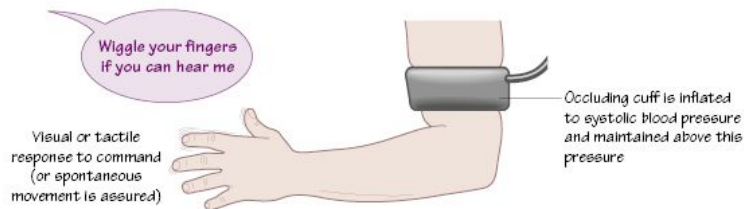
Towards a consciousness marker

- Analyze MNS-induced EEG pattern in particular anesthetic states:
 - Light sedation - paradoxical excitation^[1]
 - Deep anesthesia - burst suppression^[1]
- Verify the impact of physiological & patient-specific conditions
- **Ground truth?**
New experimental protocol combining:
 - Median Nerve Stimulation
 - Isolated Forearm Technique^[2]

Paradoxical Excitation: Beta Oscillations



Unconsciousness: Burst Suppression



[1] Purdon et al, Anesth, 2015

[2] Russel & Wang, Anaesth, 2014

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ANR BCI4IA



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Claude Meistelman



Ana Maria Cebolla
Alvarez



Guy Cheron



Philippe Guerci



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Thank you for your attention!

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